Core2350B ↔ Tang Nano20K · 16-signal wiring map

Core2350B: RP2350B / 8 MB PSRAM / 16 MB flash • Tang Nano20K hardware 3921

Bus-test build: 6.3MHz GPU; black HDMI only; electrical timing unqualified
RP2350 A2; one 100Ω per signal wire; two 4.7kΩ handshake pull-downs fitted

CORE2350B			CABLE / SIGNAL FLOW		TANG NANO20K		
GPIO	Header pin	Signal	Single series resistor	Flow	Header pin	FPGA pin	Bank
GP39	P4.12	WREADY	100Ω	←	J6.1	73	1
GP40	P4.11	RVALID	100Ω	←	J6.2	74	1
GP35	P4.8	ABORT	100Ω	→	J5.17	72	1
GP36	P4.7	GPU_RESET_N	100Ω	→	J5.18	71	1
GP37	P4.10	ALE_N	100Ω	→	J6.15	17	6
GP38	P4.9	RD_N	100Ω	→	J6.7	16	6
GP16	P3.4	HCLK	100Ω	→	J6.5	77	1
GP17	P3.3	WR_N	100Ω	→	J6.6	15	6
GP18	P3.6	D0	100Ω	↔	J6.8	27	5
GP19	P3.5	D1	100Ω	↔	J6.9	28	5
GP20	P3.8	D2	100Ω	↔	J6.12	29	5
GP21	P3.7	D3	100Ω	↔	J6.13	30	5
GP22	P3.10	D4	100Ω	↔	J6.14	31	5
GP23	P3.9	D5	100Ω	↔	J5.5	42	4
GP24	P3.12	D6	100Ω	↔	J5.6	41	4
GP25	P3.11	D7	100Ω	↔	J5.10	48	3

Power and ground

Power boards through their normal inputs.
Join common GND: Tang J6.20, J5.2, J5.15
to Core P1.1 / P3.1 (fan-out as needed).
Keep inter-board 3.3V and 5V rails separate.
Tang J6.19: local pull-up rail only.
Tang J5.1 (5V), J5.16 (3.3V): unconnected.

Bias · fitted / recommended

Fitted: 4.7kΩ each GP39 / GP40 → GND.
Added pulls: capability reads still all zero.
Other external bias parts: not confirmed.
Design: up WR_N, RD_N, ALE_N to Tang 3.3V;
down HCLK, ABORT, RESET_N to GND at Tang.
No data pulls recommended. RESET_N low = reset.

CONFIRMED assembly / shared pads

16 × 100Ω total; ONE per wire, including data.
Resistor end placement was not reported.
Disconnect the Tang FPC LCD and VGA DAC.
Preserve HDMI CEC/HPD (FPGA pins 25/26).
3 loads (15/16/17): orange LED + 510Ω to 3.3V.
RP controls 12mA; HCLK phases ≥50us.

Keep ALL 16 signal links OPEN until compatible Core firmware + FPGA image are installed.
Assembly while unpowered is OK; keep removable links open on powered old images. 100Ω does not make contention safe.

Core bus: P3/P4 contacts, not package pins. GP39 WREADY: RP input; shared red LED + 470Ω to GND adds Tang output load.

Sources: user wiring revision; Core2350B schematic; Tang Rev 1.30 / hardware 3921 (2023-07-11). Exact evidence: pinout JSON.

Tang Nano20K · physical header map

Hardware 3921 · component side · USB-C TOP / HDMI BOTTOM

Bus-test build: 6.3MHz GPU; black HDMI only; electrical timing unqualified
RP2350 A2; one 100Ω per signal wire; two 4.7kΩ handshake pull-downs fitted

J6 = LEFT edge; J5 = RIGHT edge. Both number 1 at top → 20 at bottom.



Blue: Core → Tang • Teal: Tang → Core • Purple: bidirectional data
Grey GPIOs: leave bus wiring off. Orange supply pins: never connect to a Core power rail.
Common ground: J6.20 + J5.2 + J5.15 → Core P1.1 / P3.1. Power each board separately.
FPC LCD disconnected. HDMI CEC/HPD (J6.10/J6.11) remain reserved for HDMI.
Keep ALL 16 signal links OPEN until compatible Core firmware + FPGA image are installed.
16 × 100Ω fitted (one/wire); end placement unknown. Series R does not make contention safe.

Physical orientation: view the components; do not mirror this map for the solder side. Board outline is schematic, not to scale.

Source: Tang Nano20K Rev 1.30 / hardware 3921 (2023-07-11); full provenance and signal map in the pinout JSON.

Core2350B · P4 / P3 physical locator

RP2350B package, A2 silicon · 8 MB PSRAM / 16 MB flash · component side · USB FPC TOP

Bus-test build: 6.3MHz GPU; black HDMI only; electrical timing unqualified
RP2350 A2; one 100Ω per signal wire; two 4.7kΩ handshake pull-downs fitted

Use the Core2350B connector labels below; Pico 2 header numbering does not apply.

P4 · RIGHT EDGE · vertical 2×8 header

Inner / odd		Outer / even	
P4.1 · GND Common ground to Tang	1	2	P4.2 · GP30 Unallocated by this bus layout
P4.3 · GP32 Unallocated by this bus layout	3	4	P4.4 · GP31 Unallocated by this bus layout
P4.5 · GP34 Unallocated by this bus layout	5	6	P4.6 · GP33 Unallocated by this bus layout
P4.7 · GP36 GPU_RESET_N · Tang J5.18	7	8	**P4.8 · GP35** ABORT · Tang J5.17
P4.9 · GP38 RD_N · Tang J6.7	9	10	**P4.10 · GP37** ALE_N · Tang J6.15
P4.11 · GP40 RVALID · Tang J6.2	11	12	**P4.12 · GP39** WREADY · Tang J6.1 · Core LED load
P4.13 · GP42 Unallocated by this bus layout	13	14	P4.14 · GP41 Unallocated by this bus layout
P4.15 · GP44 Unallocated by this bus layout	15	16	P4.16 · GP43 Unallocated by this bus layout

P4 contacts read TOP → BOTTOM. Outer column is toward the board's right edge.

GP39: WREADY input only; red LED + 470Ω to GND loads the Tang output. Disable RP LED writes.

P3 · BOTTOM EDGE · horizontal 2×8 header

Read both rows LEFT → RIGHT. Inner / even row is nearer the board centre.

P3.2 GP15 Unavailable	P3.4 GP16 HCLK	P3.6 GP18 D0	P3.8 GP20 D2	P3.10 GP22 D4	P3.12 GP24 D6	P3.14 GP26 Unallocated	P3.16 GP28 PS/2 kept
2	4	6	8	10	12	14	16
INNER / EVEN						OUTER / ODD	
1	3	5	7	9	11	13	15
P3.1 GND Common GND	P3.3 GP17 WR_N	P3.5 GP19 D1	P3.7 GP21 D3	P3.9 GP23 D5	P3.11 GP25 D7	P3.13 GP27 PS/2 kept	P3.15 GP29 Unallocated

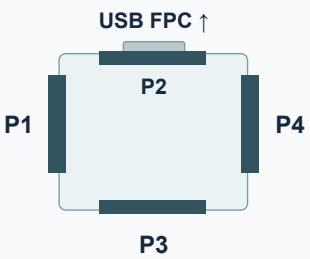
Physical orientation reference

Component side, USB FPC toward top:

P1 left · P2 top · P3 bottom · P4 right.

Signals use P3/P4. Bond P1.1 GND to Tang J5.2 as well.

Keep ALL 16 signal links OPEN until compatible Core firmware + FPGA image are installed; old powered images need links open.
One 100Ω per signal wire; end unknown. R does not make contention safe.



Connector locator only. P3/P4 drawings are expanded for legibility and are not a scale board drawing. Do not mirror for solder side.

Sources: user wiring revision, Core2350B schematic and component-side photograph; provenance in the pinout JSON.